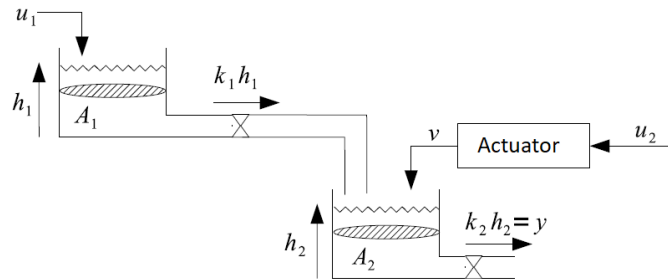


Model and Stability: Exercises

The solution of the exercises proposed in this document was given during class.

Exercise 1)

Consider the following system



Where

- $u_1(t)$ is a volume flow $\left[\frac{m^3}{min}\right]$
- $h_1(t), h_2(t)$ are the liquid levels in the tanks
- The tank area $A_1 = A_2 = 2 m^2$
- output flows $q_1(t), q_2(t)$ are governed by $q_i = k_i h_i(t)$
- The actuator is model by the following equations

$$\dot{x}_1(t) = \frac{1}{h_1 + 1} x_1(t)$$

$$\dot{x}_2(t) = -x_1(t) - x_2(t) + u_2(t)$$

$$v(t) = x_2(t)$$

Considering that we are interested in the output flow $q_2(t)$, determine the system equation.

Exercise 2)

Consider the system

$$\dot{x}_1(t) = x_1^2(t)(x_2(t) - 2) + u(t)$$

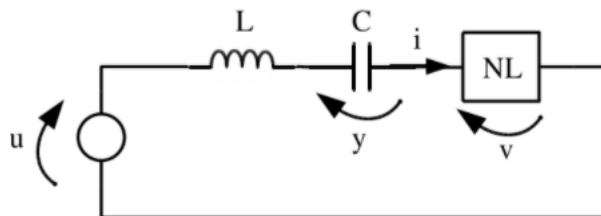
$$\dot{x}_2(t) = (x_1(t) - \alpha)(x_2(t) - 2) - x_2(t)$$

$$y(t) = x_1(t) x_2(t)$$

- Evaluate the equilibrium state connected to the input
- Determine the linearized system on the equilibrium points
- Study the stability of the equilibrium points posing $\alpha \neq -1$

Exercise 3)

Considering the following circuit:



Where $L = 1$, $C = 1$ and the element NL imposes the following relation among current and voltage

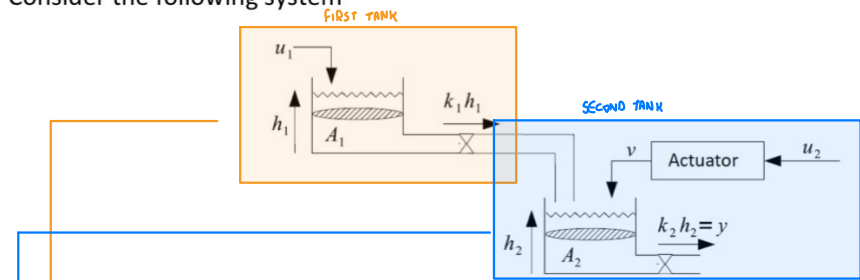
$$v(t) = i^3(t) + i(t)$$

Recalling that the tension over L and C are given by $v_L(t) = L \frac{di(t)}{dt}$ and $v_C(t) = C \int i(t) dt$ respectively.

- Determine the dynamic equation describing the system.
- Find the equilibrium states associated to the input $u(t) = \underline{u} = 1$
- Evaluate the stability of the previously defined equilibrium points

Exercise 1)

Consider the following system



Where

- $u_1(t)$ is a volume flow $\left[\frac{m^3}{min}\right]$
- $h_1(t), h_2(t)$ are the liquid levels in the tanks
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$$\dot{x}_1(t) = \frac{1}{h_1 + 1} x_1(t)$$

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$$v(t) = x_2(t)$$

Considering that we are interested in the output flow $q_2(t)$, determine the system equation.

WE ARE INTERESTED IN THE OUTPUT FLOW $q_2(t)$
DETERMINE THE SYSTEM EQUATION

STEP 1. DETERMINE INPUT, OUTPUT AND STATE

INPUT

- $u_1(t) \leftarrow$ VOLUME FLOW
- $u_2(t)$

USELESS!

STATE

- $h_1(t) = x_1(t) \leftarrow$ HEIGHT # 1
- $h_2(t) = x_2(t) \leftarrow$ HEIGHT # 2

OUTPUT

- $q_2(t)$

FIRST TANK

- $u_1(t)$
- $q_1 = k_1 h_1(t)$
- $x(t) = h_1(t)$

$$\dot{x}_1(t) = \dot{h}_1(t) = \frac{1}{A} (u_1(t) - q_1(t)) = \frac{1}{A} (u_1(t) - k_1 h_1(t))$$

SECOND TANK

USELESS!

$$\dot{x}_2(t) = \dot{h}_2(t) = \frac{1}{A} (k_1 h_1(t) + v(t) - q_2(t)) = \frac{1}{A} (k_1 h_1(t) + v(t) - k_2 h_2(t))$$

Exercise 2)

Consider the system

$\dot{x}_1(t) = x_1^2(t)(x_2(t) - 2) + u(t)$
 $\dot{x}_2(t) = (x_1(t) - \alpha)(x_2(t) - 2) - x_2(t)$
 $y(t) = x_1(t) x_2(t)$

- Evaluate the equilibrium state connected to the input $u(t)=0$
- Determine the linearized system on the equilibrium points
- Study the stability of the equilibrium points posing $\alpha \neq -1$

STATE EQUILIBRIUM

I HAVE:

$\begin{cases} \dot{x}_1(t) = x_1^2(t)(x_2(t) - 2) + u(t) \\ \dot{x}_2(t) = (x_1(t) - \alpha)(x_2(t) - 2) - x_2(t) \end{cases}$

IN EQUILIBRIUM (AKA WITH $x_1(t) = \bar{x}_1(t)$ AND $x_2(t) = \bar{x}_2(t)$) I PUT $u(t)=0$

EQUILIBRIUM
 $E_q = (\bar{x}_1(t); \bar{x}_2(t); 0)$
TO FIND

$\begin{cases} 0 = \bar{x}_1^2(t)(\bar{x}_2(t) - 2) + 0 \\ 0 = (\bar{x}_1(t) - \alpha)(\bar{x}_2(t) - 2) - \bar{x}_2(t) \end{cases}$
ALSO SET $\dot{x}_1(t)=0$ AND $\dot{x}_2(t)=0$

TO SOLVE FIRST EQ: $\bar{x}_1(t)=0 \vee \bar{x}_2(t)=2$

$\begin{cases} 0=0 \\ 0 = (\bar{x}_1(t) - \alpha) \cdot 0 - \bar{x}_2(t) \end{cases} \rightarrow \bar{x}_2(t)=0 \text{ UNSAT}$

MUST BE THIS ONE

$\begin{cases} 0=0 \\ 0 = (-\alpha)(\bar{x}_2(t) - 2) - \bar{x}_2(t) \end{cases} \rightarrow \bar{x}_2(t) + \alpha \bar{x}_2(t) - 2\alpha = 0 \rightarrow \bar{x}_2(t) = \frac{2\alpha}{1+\alpha}$

EQUILIBRIUM:
 $E_q = (0; \frac{2\alpha}{1+\alpha}; 0)$

LINEARIZED SYSTEM ON EQUILIBRIUM POINT

IF I HAVE A NON-LINEAR SYSTEM, THE LINEAR SYSTEM CAN BE OBTAINED AS FOLLOW (BY APPROXIMATION, IF WE KNOW EQ. VALUES)] THIS IS BECAUSE THE DERIVATIVE $\dot{x}(t)$ OF EQU. POINT IS 0, SO I CAN REMAIN WITH JUST THE $\dot{x}(t)$ PART

$\begin{cases} \delta u(t) = u(t) - \bar{u} \\ \delta x(t) = x(t) - \bar{x} \\ \delta y(t) = y(t) - \bar{y} \end{cases}$

$\begin{cases} \delta \dot{x}(t) = A \delta x(t) + B \delta u(t) \\ \delta \dot{y}(t) = C \delta x(t) + D \delta u(t) \end{cases}$

$\bar{y}=0$ SINCE FROM BEFORE $y(t) = x_1(t)x_2(t) \rightarrow \bar{y}(t) = \bar{x}_1(t)\bar{x}_2(t) = 0 \cdot \frac{2\alpha}{1+\alpha} = 0$

TWO X SO:

$\delta x_1(t) = x_1(t) - \bar{x}_1$

$\delta x_2(t) = x_2(t) - \bar{x}_2 = x_2(t) - \frac{2\alpha}{1+\alpha}$

$\delta u(t) = u(t) - 0$

$\delta y(t) = y(t) - 0$ SINCE $\bar{y}=0$

STUDY THE STABILITY OF THE EQUILIBRIUM POINTS POSING $\alpha \neq -1$

SINCE THE DERIVATIVE WRT A FIXED POINT IS 0, WE DETERMINE THE SYSTEM MATRICES A, B, C, D ON THE DERIVATES OF THE ORIGINAL SYSTEM EVALUATED AT THE EQUILIBRIUM POINTS

$$\begin{cases} \dot{x}(t) = Ax(t) + Bu(t) \\ y(t) = Cx(t) + Du(t) \end{cases}$$

$$\begin{aligned} \dot{x}_1(t) &= x_1^2(t)(x_2(t) - 2) + u(t) \\ \dot{x}_2(t) &= (x_1(t) - \alpha)(x_2(t) - 2) - x_2(t) \\ y(t) &= x_1(t)x_2(t) \end{aligned}$$

$$A = \begin{bmatrix} \frac{df_1(x, w)}{dx_1} & \frac{df_1(x, w)}{dx_2} \\ \frac{df_2(x, w)}{dx_1} & \frac{df_2(x, w)}{dx_2} \end{bmatrix} = \begin{bmatrix} 2x_1(x_2(t) - 2) & x_1^2(t) \\ x_2(t) - 2 & x_1(t) - \alpha - 1 \end{bmatrix} \Big|_{\bar{x}, \bar{w}}$$

$$B = \begin{bmatrix} \frac{df_1(x, w)}{du} \\ \frac{df_2(x, w)}{du} \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \Big|_{\bar{x}, \bar{w}}$$

$$C = \left[\frac{dy(x, w)}{dx_1}, \frac{dy(x, w)}{dx_2} \right] = [x_2(t); x_1(t)] \Big|_{\bar{x}, \bar{w}}$$

! THE OUTPUT FUNCTION IS y ! DO NOT USE f

$$D = \left[\frac{dy(x, w)}{dw} \right] = [0] \text{ AND NOT HERE}$$

WHY?

NOW SUBSTITUTE IN THE A, B, C, D MATRICES THE EQUILIBRIUM POINTS

$$\hookrightarrow e_2: (0, \frac{2\alpha}{\alpha+1}, 0)$$

$\uparrow$ $x_1(t)$ $\uparrow$ $x_2(t)$ $\uparrow$ $w(t)$

$$A = \begin{bmatrix} 0 & 0 \\ \frac{2\alpha}{\alpha+1} - 2 & -\alpha - 1 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$C = \left[\frac{2\alpha}{\alpha+1}; 0 \right]$$

$$D = [0]$$

TO CHECK THE STABILITY I NEED TO COMPUTE THE EIGENVALUES OF A

$$\lambda_{1,2} = \det(\lambda I - A) = \det \left(\begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} - \begin{bmatrix} 0 & 0 \\ \frac{2\alpha}{\alpha+1} - 2 & -\alpha - 1 \end{bmatrix} \right) = \det \left(\begin{bmatrix} \lambda & 0 \\ -\frac{2\alpha}{\alpha+1} + 2 & \lambda + \alpha + 1 \end{bmatrix} \right)$$

$$= \lambda^2 + \alpha\lambda + \lambda - 0 \left(-\frac{2\alpha}{\alpha+1} + 2 \right) = 0 \quad \text{SET } = 0 \text{ TO FIND } \lambda$$

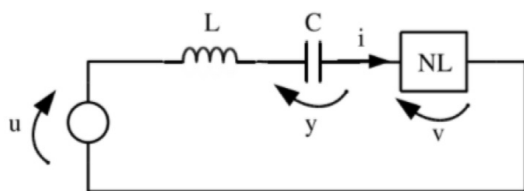
$$= \lambda(\lambda + \alpha + 1) = 0 \rightarrow \lambda_1 = 0 \vee \lambda_2 = -1 - \alpha$$

1. An LTI system is **asymptotically stable** if every eigenvalue has a real part < 0
2. An LTI system is **unstable** if at least one of the eigenvalues has a **strictly** positive real part
3. If the eigenvalues of a LTI system are **not** positive, the system can be stable or unstable but it **cannot** be asymptotically stable
4. If the eigenvalues are along the imaginary axis, i.e., when the real part is 0, the system is **stable**, but **not** asymptotically (i.e., "simply stable")

- SINCE $\lambda_1 = 0$ AND IS NOT < 0 THE SYSTEM CAN NOT BE ASYMPTOTICALLY STABLE
- IF $\alpha > -1$ THEN $\lambda_2 > 0 \rightarrow$ UNSTABLE
- IF $\alpha < -1$ THEN $\lambda_2 < 0 \rightarrow$ SIMPLY STABLE

Exercise 3)

Considering the following circuit:



Where $L = 1$, $C = 1$ and the element NL imposes the following relation among current and voltage

$$v(t) = i^3(t) + i(t)$$

Recalling that the tension over L and C are given by $v_L(t) = L \frac{di(t)}{dt}$ and $v_C(t) = C \int i(t) dt$ respectively.

- Determine the dynamic equation describing the system.
- Find the equilibrium states associated to the input $u(t) = \underline{u} = 1$
- Evaluate the stability of the previously defined equilibrium points

① DYNAMIC EQUATION

$$v_L(t) = L \frac{di(t)}{dt} = \frac{d: (t)}{dt}$$

$$v_C(t) = C \int i(t) dt = \int i(t) dt$$

KIRCHHOFF'S VOLTAGE LAW

SUM OF ALL VOLTAGES IN A CLOSE LOOP MUST BE ZERO. THE SOURCE $w(t)$ IS THE SUM OF VOLTAGES ACROSS L , C AND NL

$$w(t) = v_L(t) + v_C(t) + v_{NL}(t) = \frac{d: (t)}{dt} + \int i(t) dt + i^3(t) + i(t)$$

Why

$$v_C(t) = x_2(t)$$

REMOVE THE INTEGRAL

$$v_C(t) = \int i(t) dt \rightarrow \frac{d}{dt} v_C(t) = \dot{v}_C(t) = i(t) = x_2(t)$$

$$v_L(t) = \frac{d: (t)}{dt} = \frac{d \dot{v}_C(t)}{dt} = \frac{d x_2(t)}{dt}, \quad \dot{x}_2(t) = x_3(t)$$

$$\begin{cases} x_1(t) = v_C(t) \\ x_2(t) = \dot{v}_C(t) \\ x_3(t) = v_L(t) \\ y = v(t) \end{cases} \quad \begin{cases} \dot{x}_1(t) = x_2(t) \\ \dot{x}_2(t) = x_3(t) \\ \dot{x}_3(t) = -x_1(t) - x_2^3(t) - x_3(t) + w(t) \\ y(t) = x_2^3(t) + x_2(t) \end{cases}$$

REWRITE

② EQUILIBRIUM STATES ASSOCIATED TO THE INPUT $w(t) = \underline{u} = 1$

$$eq((\bar{x}_1, \bar{x}_2), 1)$$

BECAUSE TEXT TELL US THAT $\underline{u} = 1$

$$\begin{cases} 0 = x_2(t) \\ 0 = -x_1(t) - x_2^3(t) - x_2(t) + 1 \end{cases} \quad \begin{cases} 0 = x_2(t) \\ 0 = -x_1(t) - 0 - 0 + 1 \end{cases} \rightarrow x_1(t) = 1 \quad \rightarrow eq((1, 0), 1)$$

(3) STABILITY of EQUILIBRIUM POINT

$$\begin{cases} \dot{x}_1(t) = x_2(t) \\ \dot{x}_2(t) = -x_1(t) - x_2^3(t) + w(t) \\ y(t) = x_2^3(t) + x_2(t) \end{cases}$$

$$\begin{cases} \dot{x}(t) = Ax(t) + Bw(t) \\ \dot{y}(t) = Cx(t) + Dw(t) \end{cases}$$

$$A = \begin{bmatrix} \frac{df_1(x,w)}{dx_1} & \frac{df_1(x,w)}{dx_2} \\ \frac{df_2(x,w)}{dx_1} & \frac{df_2(x,w)}{dx_2} \end{bmatrix} \bigg|_{x,w} = \begin{bmatrix} 0 & 1 \\ -1 & -3x_2^2(t) - 1 \end{bmatrix} \bigg|_{x,w} = \begin{bmatrix} 0 & 1 \\ -1 & -1 \end{bmatrix}$$

Eigen $((1, 0), 1)$
 $\lambda_2 = 0$
 -1

$$B = \begin{bmatrix} \frac{df_1(x,w)}{dw} \\ \frac{df_2(x,w)}{dw} \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$C = \begin{bmatrix} \frac{dy(x,w)}{dx_1} & \frac{dy(x,w)}{dx_2} \end{bmatrix} \bigg|_{x,w} = \begin{bmatrix} 0 & 3x_2^2(t) + 1 \end{bmatrix} \bigg|_{x,w} = \begin{bmatrix} 0 & 1 \end{bmatrix}$$

$$D = \begin{bmatrix} \frac{dy(x,w)}{dw} \end{bmatrix} = \begin{bmatrix} 0 \end{bmatrix}$$

USE FOR EIGENVALUES

$$\lambda_{1,2} = \text{DET}(\lambda I - A) = \text{DET} \begin{pmatrix} \lambda & -1 \\ 1 & \lambda + 1 \end{pmatrix} = \lambda^2 + \lambda - (-1) = \lambda^2 + \lambda + 1$$

$$\lambda_1 = \frac{-1 + \sqrt{1-4}}{2} = \frac{-1 + i\sqrt{3}}{2}$$

$$\lambda_2 = \frac{-1 - \sqrt{1-4}}{2} = \frac{-1 - i\sqrt{3}}{2}$$

$\text{Re} < 0, \text{Im} \neq 0 \rightarrow \text{ASYMPTOTIC STABLE}$